NIOS Class 10 (211) Mathematics — Master Formula Cheat Sheet

NIOS Secondary (Class 10) Mathematics (211) — Exam Prep Study Kit

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1-Stop Formula Reference for Public Examination

> **How to use this**: Keep this file open on your phone or print it out. Review these formulas every single day until 5th November. Over 60% of exam marks come from direct application of these formulas!

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1. Trigonometry & Heights and Distances (Module 5 — 10 Marks)

Trigonometric Values Table (MEMORIZE THIS ENTIRE ROW!)

Angle (θ)	0°	30°	45°	60°	90°
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not defined
$\operatorname{cosec}\theta$	Not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec\theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not defined
$\cot\theta$	Not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

> **Pro Tip for Heights & Distances (5 Marks Q44)**: In almost every problem, you will use $\tan\theta = \frac{\text{Height (Opposite)}}{\text{Distance (Adjacent)}}$: - When $\theta = 45^\circ \implies \tan 45^\circ = 1 \implies \text{Height} = \text{Distance}$! - When $\theta = 30^\circ \implies \tan 30^\circ = \frac{1}{\sqrt{3}}$ - When $\theta = 60^\circ \implies \tan 60^\circ = \sqrt{3}$ - Remember: $\sqrt{3} \approx 1.732$.

Core Trigonometric Identities

- $\sin^2\theta + \cos^2\theta = 1 \implies \sin^2\theta = 1 - \cos^2\theta \implies \cos^2\theta = 1 - \sin^2\theta$
- $1 + \tan^2\theta = \sec^2\theta \implies \sec^2\theta - \tan^2\theta = 1$
- $1 + \cot^2\theta = \text{cosec}^2\theta \implies \text{cosec}^2\theta - \cot^2\theta = 1$
- $\tan\theta = \frac{\sin\theta}{\cos\theta}, \quad \cot\theta = \frac{\cos\theta}{\sin\theta}, \quad \sec\theta = \frac{1}{\cos\theta}, \quad \text{cosec}\theta = \frac{1}{\sin\theta}$

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2. Algebra (Module 1 — 20 Marks)

Special Products & Factorisation (Ch 4)

- $(a+b)^2 = a^2 + 2ab + b^2$
- $(a-b)^2 = a^2 - 2ab + b^2$
- $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$
- $(a+b)^2 - (a-b)^2 = 4ab$
- $a^2 - b^2 = (a+b)(a-b)$
- $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$
- $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$
- $(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$

Linear Equations in Two Variables (Ch 5)

For system $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$: - $\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \implies$ **Unique solution** (Lines intersect at 1 point, Consistent). - $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \implies$ **Infinitely many solutions** (Coincident lines, Dependent Consistent). - $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies$ **No solution** (Parallel lines, Inconsistent).

Quadratic Equations (Ch 6)

Standard form: $ax^2 + bx + c = 0$ $(a \neq 0)$ - **Discriminant**: $D = b^2 - 4ac$ - **Nature of Roots**: - $D > 0 \implies$ Two real and distinct roots - $D = 0 \implies$ Two real and equal roots ($x = -\frac{b}{2a}$) - $D <$

$0 \implies$ No real roots - **Quadratic Formula**:

$$x = \frac{-b \pm \sqrt{D}}{2a} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- **Sum & Product of Roots**: - Sum of roots $\alpha + \beta = -\frac{b}{a}$ - Product of roots $\alpha \cdot \beta = \frac{c}{a}$

Arithmetic Progression (AP - Ch 7)

- Common difference: $d = a_n - a_{n-1}$
- n**-th term**:

$$a_n = a + (n-1)d$$

- **Sum of first** n **terms**:

$$S_n = \frac{n}{2}[2a + (n-1)d] \quad \text{or} \quad S_n = \frac{n}{2}[a + l]$$

(where $l = a_n$ *is the last term)*

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3. Coordinate Geometry (Module 3 — Ch 19 — 6 to 8 Marks)

- **Distance between two points** $P(x_1, y_1)$ **and** $Q(x_2, y_2)$:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- **Distance of point** (x, y) **from origin** $(0, 0)$:

$$d = \sqrt{x^2 + y^2}$$

- **Mid-point of line segment joining** (x_1, y_1) **and** (x_2, y_2):

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

- **Section Formula (Internal division in ratio** $m : n$**)**:

$$P(x, y) = \left(\frac{mx_2 + nx_1}{m + n}, \frac{my_2 + ny_1}{m + n} \right)$$

(Tip: If point is on y-axis, $x = 0$*. If point is on x-axis,* $y = 0$*)*

- **Centroid of a Triangle with vertices** $(x_1, y_1), (x_2, y_2), (x_3, y_3)$:

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

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4. Circle Geometry & Theorems (Module 3 — Ch 16 & 17 — 10 Marks)

1. **Tangent $\perp$ Radius**: A tangent at any point of a circle is perpendicular to the radius through the point of contact ($\angle OPT = 90^\circ$).
2. **Lengths of Tangents**: The lengths of two tangents drawn from an external point to a circle are equal ($PA = PB$).
3. **Angle Subtended at Centre**: The angle subtended by an arc at the centre is twice the angle subtended by it at any point on the remaining part of the circle ($\angle AOC = 2\angle ABC$).
4. **Angle in Semicircle**: Angle in a semicircle is a right angle (90°).
5. **Angles in Same Segment**: Angles in the same segment of a circle are equal.
6. **Cyclic Quadrilateral**: The sum of either pair of opposite angles of a cyclic quadrilateral is 180° ($\angle A + \angle C = 180^\circ,\ \angle B + \angle D = 180^\circ$).

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5. Mensuration (Module 4 — Ch 21 — 10 Marks)

Solid Shape	Curved Surface Area (CSA)	Total Surface Area (TSA)	Volume (V)
Cuboid (l, b, h)	$2h(l + b)$ *(Area of 4 walls)*	$2(lb + bh + hl)$	$l \times b \times h$
Cube (side a)	$4a^2$	$6a^2$	a^3
Cylinder (radius r, height h)	$2\pi rh$	$2\pi r(r + h)$	$\pi r^2 h$
Cone (radius r, height h, slant l)	πrl *(where $l = \sqrt{r^2 + h^2}$)*	$\pi r(r + l)$	$\frac{1}{3}\pi r^2 h$
Sphere (radius r)	$4\pi r^2$	$4\pi r^2$	$\frac{4}{3}\pi r^3$
Hemisphere (radius r)	$2\pi r^2$	$3\pi r^2$	$\frac{2}{3}\pi r^3$

- **Circle Sector & Ring**:

- Area of circle $= \pi r^2$, Circumference $= 2\pi r$
- Area of Sector with central angle $\theta = \frac{\theta}{360^\circ} \times \pi r^2$
- Length of Arc $= \frac{\theta}{360^\circ} \times 2\pi r$
- Area of Circular Ring $= \pi(R^2 - r^2) = \pi(R + r)(R - r)$

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6. Statistics & Probability (Module 6 — Ch 25 & 26 — 12 Marks)

Statistics (Ch 25)

- **Mean (Direct Method)**:

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

- **Median of Grouped Data (3 Marks Sure-Shot!)**:

$$\text{Median} = L + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h$$

- L = Lower limit of median class
- $N = \sum f_i$ = Total frequency
- cf = Cumulative frequency of the class *preceding* the median class
- f = Frequency of the median class
- h = Class width (upper limit − lower limit)
- **Empirical Formula (Very common 1-mark question!)**:

$$\text{Mode} = \mathbf{3}\,\text{Median} - \mathbf{2}\,\text{Mean}$$

Probability (Ch 26)

- $$P(E) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

- Range of probability: $0 \leq P(E) \leq 1$
- Sum of complementary probabilities: $P(E) + P(\bar{E}) = 1 \implies P(\text{not } E) = 1 - P(E)$
- Standard coin toss outcomes:
- 1 coin: $\{H, T\}$ (Total = 2)
- 2 coins: $\{HH, HT, TH, TT\}$ (Total = 4)
- Standard dice outcomes:

- 1 die: $\{1, 2, 3, 4, 5, 6\}$ (Total = 6)
- 2 dice: Total outcomes = $6 \times 6 = 36$
- Standard deck of cards: Total = 52 cards (26 Red: 13 Hearts + 13 Diamonds; 26 Black: 13 Spades + 13 Clubs; 12 Face Cards: 4 Kings + 4 Queens + 4 Jacks; 4 Aces).

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7. Commercial Mathematics — Installment Buying (Ch 9 — 8 Marks)

- **Cash Price**: Price of the article if paid in full immediately.
- **Cash Down Payment (CDP)**: Amount paid in cash at the time of purchase.
- **Unpaid Balance (Principal)** $= \text{Cash Price} - \text{Cash Down Payment}$
- **Total Amount Paid under Installment Scheme** $= \text{Down Payment} + (n \times \text{Monthly Installment Amount})$
- **Total Interest Paid** $= \text{Total Amount Paid} - \text{Cash Price}$